Prototyping a Weather Radar Foundation Model Using a Self-Supervised Learning Architecture

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ABSTRACT

In meteorology, machine learning is emerging as a leading solution to many important problems. However, many commonly used methods rely on supervised learning, which requires large labeled datasets that are often created through time-intensive manual annotation. The limited availability of these datasets represents a substantial barrier to the development of broadly applicable machine-learning tools for the weather enterprise. In this study, we investigate image-based self-supervised learning as an avenue for pretraining a general-purpose model using polarimetric observations from the KHTX S-band Next Generation Weather Radar (NEXRAD) site. The proposed framework adapts Meta AI's self-distillation with no labels (DINO) architecture for multichannel meteorological imagery through the implementation of a custom channel-wise tokenizer. This tokenizer independently encodes information from each polarimetric radar variable before integrating the resulting representations within a shared vision-transformer framework, allowing the model to preserve both variable-specific information and spatial relationships among radar features. Rather than learning from predefined meteorological labels, the model is trained to produce meaningful feature embeddings directly from the spatial, textural, and cross-variable characteristics of radar observations. These learned representations provide a potential foundation for downstream models that could be adapted to specific meteorological applications using substantially smaller labeled datasets. This framework may reduce dependence on extensive manual labeling while improving the transferability of learned representations across weather phenomena, radar sites, and downstream tasks. More broadly, this work evaluates the potential of DINO-based self-supervised pretraining to support scalable foundation-model development for polarimetric radar analysis and operational meteorological applications.

1. Introduction

Weather radar provides detailed observations of precipitation structure and storm evolution, while polarimetric variables offer additional information about hydrometeor characteristics and storm organization. These observations support applications such as precipitation estimation, storm classification, and severe-weather detection. However, many radar-based machine-learning methods rely on supervised learning and therefore require large, manually labeled datasets. Creating these labels is time-intensive, often requires expert meteorological knowledge, and limits the range of phenomena available for model development.

Self-supervised learning provides an alternative by allowing models to learn useful representations directly from unlabeled data. The DINO framework trains student and teacher vision transformers to produce similar representations from differently augmented views of the same image (Caron et al. 2021). Previous work has shown that this approach can extract meaningful atmospheric features from unlabeled cloud-camera imagery (Dematties et al. 2023), suggesting that similar methods may be useful for weather-radar observations.

Applying DINO to polarimetric radar data requires accounting for variables with different physical meanings, numerical ranges, and missing-data characteristics. This study develops a prototype radar foundation model using reflectivity, specific differential phase, differential reflectivity, cross-correlation ratio, and spectrum width from the KHTX NEXRAD site. A channel-aware tokenizer processes each radar field independently before combining the resulting representations within a shared vision transformer. The learned embeddings are evaluated using clustering, self-organizing maps, and attention-map visualization. The primary objective is to determine whether the model can identify meaningful storm structures without receiving storm-type or hazard labels during training.

2. Methods

This study explores a self-supervised learning workflow designed to characterize scenes of polarimetric weather-radar data without using manually assigned meteorological labels in the pretraining objective. The workflow begins with data acquisition and standardization of radar observations, followed by physical preprocessing, model training, feature extraction, clustering, and visual interpretation.

Data Acquisition and Processing

To support collaboration with the ARM Bankhead National Forest observatory, KHTX radar observations providing regional coverage of northern Alabama were selected for training. These data were sourced from the KHTX S-band radar near Huntsville, Alabama. Using the Adapt processing pipeline, raw NEXRAD Level-II volume scans were retrieved from the AWS-hosted NEXRAD archive. To maximize the amount of meteorologically meaningful information in the training dataset, events within a 3-hour window of a National Weather Service storm report were selected.

The storm-report categories considered were blizzard, flood, flash flood, funnel cloud, hail, heavy rain, heavy snow, lightning, sleet, tornado, thunderstorm wind, winter storm, and tropical storm. These categories were selected to increase the likelihood that convective or stratiform precipitation structures appeared in the radar scans. Most of the resulting storm reports were associated with thunderstorm wind, followed by tornado reports.

After the files were downloaded, only samples with more than 10% coverage of reflectivity exceeding 10 dBZ were retained. The radar volumes were processed into NetCDF files and regridded from the native radial coordinates to a Cartesian spatial grid. Additionally, the CSU_RadarTools package was used to estimate specific differential phase from the differential-phase observations in each radar scan.

Model Configuration

During training, the data loader selects the first available time index and extracts a horizontal radar slice nearest 2 km altitude. This height was selected to emphasize storm-scale horizontal structures while reducing the influence of near-surface clutter and strong height-dependent sampling differences.

The final field-token configuration used reflectivity, specific differential phase, differential reflectivity, cross-correlation ratio, and spectrum width. Each field was clipped to physically motivated limits before normalization to the interval $[0, 1]$. The clipping ranges were 10–75 dBZ for reflectivity, -5 to 10 deg km^{-1} for specific differential phase, -8 to 12 dB for differential reflectivity, 0.2–1.05 for cross-correlation ratio, and 0–20 m s^{-1} for spectrum width.

Values outside these ranges were clipped, and missing values were represented using a sentinel value of -1.0. Where reflectivity was missing or below the retained 10-dBZ threshold, all radar fields were assigned the missing-data sentinel. This approach retained the spatial footprint of radar coverage while distinguishing missing or excluded regions from valid normalized observations.

The model was based on the DINO self-distillation framework developed by Caron et al. (2021). The repository was adapted from the cloud-photography application developed by Dematties et al. (2023), but was modified to accept polarimetric radar variables rather than the red, green, and blue channels of a conventional image.

The primary architectural modification was a channel-aware tokenizer similar to the Channel Vision Transformer developed by Bao et al. (2024). Rather than mixing all radar variables at the image input, the model tokenized each field independently using a shared patch projection.

A learned "field" embedding was added to each token, allowing the model to preserve both the physical identity of the radar variable and the spatial location of each patch.

In the final model configuration, the 1-km gridded radar data were represented using 300 km $\times$ 300 km global views and 100 km $\times$ 100 km local views. Each view was divided into patches measuring 10 km $\times$ 10 km. The global views contained spatial information from all five radar variables represented by field-specific tokens. This design allowed the model to learn patterns within the radar structures without disregarding the physical distinctions among reflectivity, differential phase, and the other polarimetric variables.

The radar data were augmented to prevent the model from simply memorizing individual scans and instead encourage it to learn the underlying structures of radar scenes. For each input scan, two global crops and four local crops were generated. The teacher network received only the two global crops, whereas the student network received all six views. Each view had a 50% probability of being flipped horizontally or vertically. The second global crop and all local crops also had a 50% probability of receiving Gaussian noise.

Finally, one randomly selected radar field was completely masked in every student input, encouraging the model to use relationships among the remaining radar fields. The student network was trained to match the teacher's feature embeddings across different views of the same radar scene. The teacher's weights were initialized from the student network and subsequently updated using an exponential moving average of the student's parameters.

Using PyTorch, training was performed in parallel across several GPUs. Checkpoints stored the student and teacher model states and the additional training information needed to resume an interrupted run. Standard inference produced a feature vector for each radar scan along with an attention-map visualization. Associative inference was not used to produce attention maps, but it retained the matching filename and source path to support future analysis.

Post-processing

The extracted features were analyzed to determine whether the learned embeddings represented meaningful meteorological structures. Several dimensionality-reduction and unsupervised-clustering approaches were tested, including principal component analysis (PCA) followed by HDBSCAN clustering. UMAP and t-SNE projections combined with DBSCAN clustering were also evaluated. A 2×2 self-organizing map (SOM), selected based on a local minimum in topographic error, became the primary focus of the analysis. Visual examination of the SOM nodes suggested that the learned embeddings separated radar scenes according to aspects of storm organization.

3. Results

SOM Clustering

After clustering the raw feature vectors using a self-organizing map, several recurring patterns were identified within the resulting nodes. Figure 2 shows a representative case for each node. Each SOM node is represented by a learned prototype, or codebook, vector in the feature-embedding space. For each node, the cosine distance between its prototype vector and each assigned sample was calculated, and the sample with the smallest distance was selected as the representative case.

Nodes 00, 01, and 11 primarily show cellular structures, with nodes 00 and 01 having slightly more stratiform components than node 11. Node 11 shows an almost purely cellular storm mode, while node 10 shows a primarily stratiform or organized regime.

Using the labeling tool in scikit-image, 30-dBZ objects greater than 10 km^2 in area were identified as a simple proxy for discrete moderate-to-high-reflectivity features. As shown in Fig. 3, node 10 contains the fewest identified reflectivity objects, while node 11 contains the most.

The fractional coverage of the largest object was also analyzed to determine whether each radar scene was dominated by a single object. As shown in Fig. 4, node 10 primarily contains scenes dominated by one large object, whereas the other nodes show more fragmented or cellular storm structures.

Attention Maps

Transformer attention maps visualize how strongly tokens are weighted relative to one another within a selected layer and attention head. These maps can help identify spatial patterns associated with the model's internal representations, although they should not be interpreted as direct measures of causal feature importance.

In the cloud-image application of Dematties et al. (2023), individual attention heads produced recognizable feature masks, with different heads focusing on clouds, the Sun, the fisheye lens, and other image components. Similar attention plots were produced for the radar model, although comparably distinct physical interpretations were not identified. Overall, attention head 1 consistently assigned greater attention to convective regions, while the remaining attention heads were less consistent.

4. Discussion

Although this prototype was limited in scope by the 10-week duration of the project, it was successful as a feasibility study for a weather-radar foundation model. We reproduced much of the cloud-image workflow developed by Dematties et al. (2023), but used a dataset composed of radar NetCDF files rather than cloud-camera images.

The primary limitations of the project were twofold. First, the model was trained on only approximately 8,500 radar scenes because the data-download and regridding pipeline was timeconsuming. Foundation models are generally trained on substantially larger and more diverse datasets. Second, the 10-km patch size may have caused the model to miss meaningful structures within convective systems. Smaller patches were not used because their substantially greater memory requirements exceeded the available GPU capacity.

There was also insufficient time to evaluate the model through downstream tasks. Nevertheless, the organization of the SOM nodes suggests that the embeddings contain information related to radar-observed storm morphology.

Several attention heads assigned substantial weight to missing-data regions. This behavior may be related to the spatially consistent sentinel value, although an ablation experiment would be required to establish the cause.

5. Conclusion

The primary conclusion of this project is that DINO could provide the basis for a weather-radar foundation model. However, significant work remains before this goal can be achieved. Smaller patches, potentially less than 5 km, may allow the model to represent finer-scale convective structures, although this hypothesis requires direct testing. Training on a much larger and more diverse dataset may also improve representation quality and generalization.

Alternative treatments of missing data should be tested to determine whether they produce more physically meaningful attention patterns. Future work should also evaluate the pretrained embeddings through downstream classification, segmentation, or retrieval tasks and test whether the learned representations transfer across radar sites and meteorological regimes.

If these developments are successful, this project could provide significant value to the broader meteorological community. Publicly releasing the pretrained model weights could help democratize radar-based machine-learning research by providing a reusable foundation for lighter-weight, targeted downstream tasks.

References

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Dematties, D., B. A. Raut, S. Park, R. C. Jackson, S. Shahkarami, Y. Kim, R. Sankaran, P. Beckman, S. M. Collis, and N. Ferrier, 2023: Let's unleash the network judgment: A self-supervised approach for cloud image analysis. *Artif. Intell. Earth Syst.*, **2**, e220063, https://doi.org/10.1175/AIES-D-22-0063.1.

LIST OF FIGURES

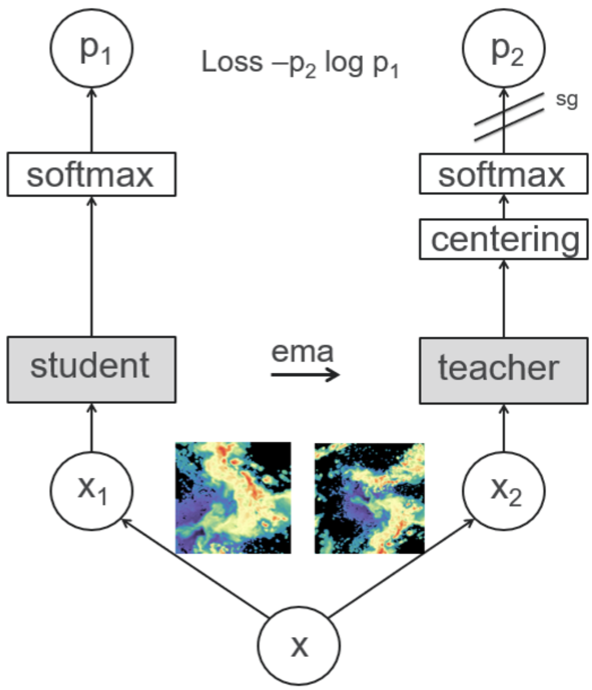

FIG. 1. Experimental model configuration.

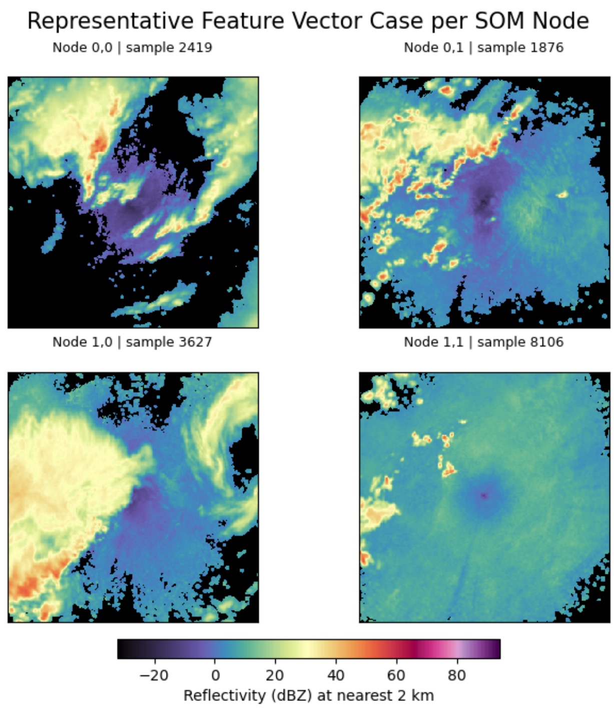

FIG. 2. Representative cases from the 2×2 SOM.

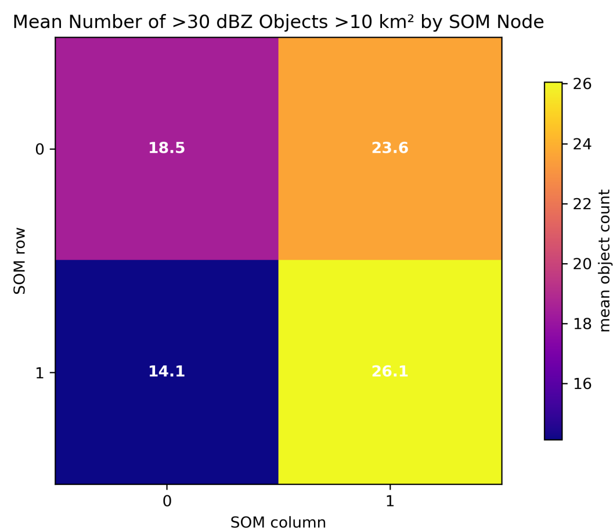

FIG. 3. Mean number of reflectivity objects exceeding 30 dBZ and 10 km^2 within each SOM node.

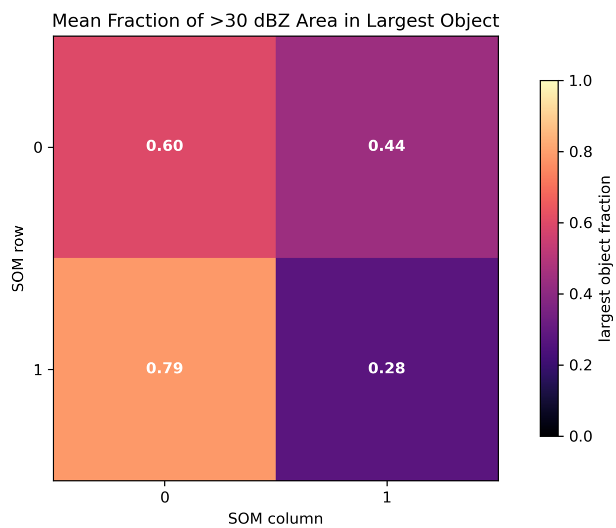

FIG. 4. Fraction of each radar scene occupied by its largest reflectivity object within each SOM node.

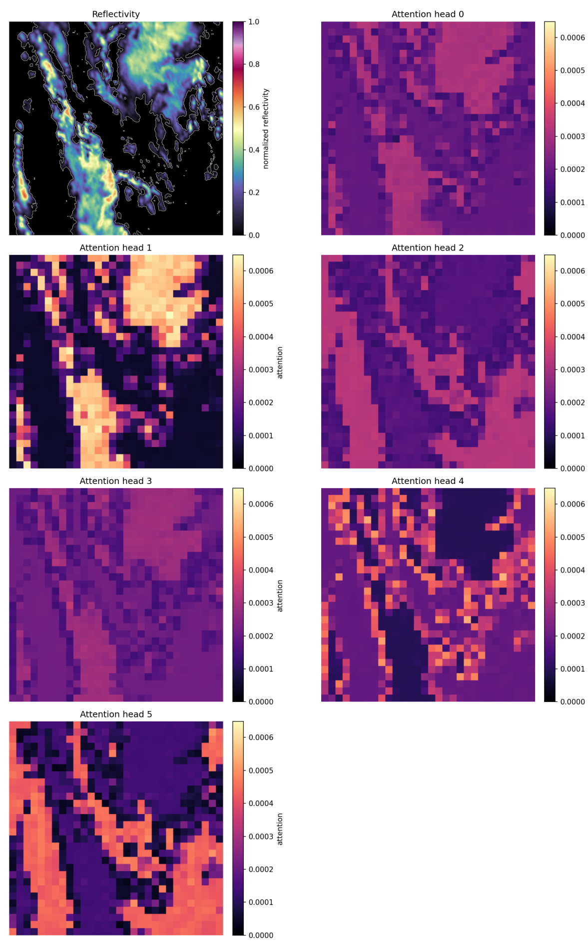

FIG. 5. Attention heads produced for an example radar sample.